ANSWER KEY & MARKING SCHEME SET A

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 Hz	oscillations ÷ time
2	**(d)** 0.2 s	T = 1/ν
3	**(a)** 340 m s^{-1}	v = ν λ
4	**(d)** 0.5 m	λ = v/ν
5	**(a)** 34 m	distance = vt/2
6	**(d)** 3812.5 m	d = vt/2
7	**(b)** vacuum	needs a medium
8	**(c)** infrasound	below 20 Hz
9	**(b)** solids	closely packed particles
10	**(d)** reverberation	gap < 0.05 s

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sound is produced by vibrating objects (1); a plucked rubber band or a tuning fork (1).	1+1
12	There is no medium in space and sound cannot travel in vacuum (2).	2
13	Compression: region of higher density; rarefaction: region of lower density in a medium (1+1).	1+1
14	The particles of the medium vibrate parallel to the direction of propagation (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stretch a rubber band over a box and pluck it: vibrations are seen and sound is heard (1.5); when the vibration stops, the sound stops (1.5).	1½+1½
16	An electric bell in a jar; air is removed by a pump (1); sound becomes fainter and almost inaudible (1); sound needs a medium (1).	1+1+1
17	The piston compresses and rarefies air in turn (1); the disturbance moves on through collisions (1); particles only oscillate about their mean positions (1).	1+1+1
	OR: Longitudinal: particle vibration parallel to propagation, e.g. sound (1.5); transverse: perpendicular, e.g. light (1.5).	1½+1½
18	Density against distance curve (1); highest point crest, lowest trough (1); distance between two crests is the wavelength (1).	1+1+1
	OR: ν = 10/2 = 5 Hz (1.5); T = 1/5 = 0.2 s (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Plucked rubber band on a box (2). (ii) Bell jar experiment (2). (iii) No medium in space (1).	2+2+1
	OR: (i) Piston model: alternate compressions and rarefactions (2). (ii) No; particles oscillate about their positions and only energy moves (1). (iii) Parallel vs perpendicular vibration (2).	2+1+2
20	(i) T = 1/256 s ≈ 0.0039 s (1). (ii) λ = 344/256 ≈ 1.34 m (2). (iii) 1280 (1). (iv) λ = 1500/256 ≈ 5.86 m, larger (1).	1+2+1+1
	OR: (i) Four definitions (2). (ii) Larger amplitude is louder; higher frequency, higher pitch (2). (iii) 20 Hz to 20 kHz (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 Hz	oscillations ÷ time
2	**(c)** 0.05 s	T = 1/ν
3	**(b)** 340 m s⁻¹	v = ν λ
4	**(b)** 3 m	λ = v/ν
5	**(a)** 510 m	distance = vt/2
6	**(d)** 1020 m	d = v × t
7	**(b)** longitudinal waves	particles vibrate parallel to propagation
8	**(b)** 20 Hz to 20 000 Hz	average healthy ear
9	**(a)** decibel	dB
10	**(a)** echolocation	reflected ultrasonic waves

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	λ: distance between two consecutive crests; ν: oscillations per second; T: time for one oscillation (2).	2
12	T = 1/ν (1); T = 0.2 s (1).	1+1
13	v = ν λ (2).	2
14	v = 200 × 1.7 = 340 m s⁻¹ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Speed = distance/time = λ/T (2); since ν = 1/T, v = ν λ (1).	2+1
16	λ = 344/20 = 17.2 m (1.5); λ = 344/20 000 = 0.0172 m = 1.72 cm (1.5).	1½+1½
17	Light arrives almost instantly (1); d = 340 × 5 = 1700 m (2).	1+2
	OR: Fastest in solids, slower in liquids, slowest in gases (2); increases with temperature (1).	2+1
18	Sound goes and returns, so d = 340 × 0.5/2 = 85 m (3).	3
	OR: Time gap 0.1 s; total distance 340 × 0.1 = 34 m; to and fro, so 17 m (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 17.2 m (1). (ii) 1.72 cm (1). (iii) Below 20 Hz is infrasonic; above 20 kHz ultrasonic (1). (iv) Elephants hear infrasound; bats hear ultrasound (2).	1+1+1+2
	OR: (i) v = λ/T = νλ (2). (ii) 340 m s⁻¹ (1). (iii) The frequency stays the same but the speed changes (2).	2+1+2
20	(i) d = 1525 × 2.5 = 3812.5 m (2). (ii) Ultrasound travels in narrow beams with little spreading (1). (iii) Reflection of sound (1). (iv) 1525 m (1).	2+1+1+1
	OR: (i) Reflected sound heard after the original, gap ≥ 0.1 s (1). (ii) 17 m (1). (iii) Reverberation has gaps < 0.05 s; it is repeated reflections (1). (iv) Absorbing panels, curtains, upholstered seats (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.